Exercise 02: Substitution Method

Write a text file named '`substitution.txt`' that shows each recurrence has the given runtime using the substitution method.

Recurrences

1. Show $T(n) = T(\lfloor n/2 \rfloor) + 1$ is $O(\lg(n))$
2. Show $T(n) = 2T(\lceil n/2 \rceil) + n$ is $O(n \lg(n))$
3. Show $T(n) = T(n-1) + n$ is $O(n^2)$
4. Show $T(n) = T(\sqrt{n}) + 2$ is $O(\sqrt{n} \lg(n))$